QUANT

Algebra - 1

(Linear, Quadratic Equations & other imp. concepts)

Assignment 03

hitbullseye.com

1. Al is a commercial artist who uses colored pencils to sketch. How many pencils does Al have if all of them are blue except 2, all of them are yellow except 2, and all of them are red except 2?

1. 2 2. 3 3. 4 4. 6

2. Defined that $x \# y = x^2 + y^2 - xy$, then $4 \# (5 \# 6) =$

1. 663 2. 763 3. 853 4. 653

3. If $a + b = 14$, $ab = 20$, find the value of $a^3 + b^3$.

1. 2004 2. 1804 3. 2744 4. 1904

4. Defined that $z \# x = z^2 - x^2$, then if $P \# 12 = 1012$, $P =$

1. 1156 2. 34 3. 32 4. 46

5. If $a + b + c = 3$; $a^2 + b^2 + c^2 = 6$ and $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$, where a, b, c are all non-zero, then 'abc' is

1. 3 2. $\frac{1}{3}$ 3. $\frac{3}{2}$ 4. $\frac{2}{3}$

6. Find the value of $\sqrt{\sqrt{3+2\sqrt{2}} + \sqrt{11-6\sqrt{2}}}$.

1. 1 2. 2 3. 3 4. 4

7. If $2^{x+1}.5^x = 200$, what is the value of x?

1. -1 2. -2 3. 4 4. 2

8. If x_1 and x_2 are the roots of the quadratic equation $ax^2 + bx + c = 0$, find the equation whose roots are $\frac{1}{x_1}$ *and* $\frac{1}{x_2}$.

1. $cx^2 + bx + a = 0$ 2. $cx^2 + bx + c = 0$ 3. $cx^2 - bx + c = 0$ 4. $cx^2 - bx - a = 0$

9. Arjun is twice as old as Shriya. Five years ago his age was three times Shriya's age. Find the present ages of Shriya and Arjun respectively.

1. 10, 20 2.20,10 3.12, 24 4.8, 24

10. A positive number is 5 times another number. If 21 is added to both the numbers, then one of the new numbers becomes twice the other new number. If we add one to the square of the smaller number the result would be

1. 26 2. 50 3.37 4. 64

Answer key & Explanations

1.	2	Total number of pencils = (2 + 2 + 2)/2 = 3.
2.	3	$x \# y = x^2 + y^2 - xy$, $4 \#(5 \# 6) = 4 \# [5^2 + 6^2 - 5 \times 6]$ $= 4 \#[25 + 36 - 30] = 4 \# 31 = [4^2 + 31^2 - 4 \times 31]$ $= 16 + 961 - 124 = 853$.
3.	4	$a + b = 14$, $ab = 20$, $a^3 + b^3 = (a + b)(a^2 + b^2 - ab)$. $(a + b)^2 = a^2 + b^2 + 2ab$. $\therefore 196 = a^2 + b^2 + 40$ $\Rightarrow a^2 + b^2 = 156$ $\therefore a^3 + b^3 = 14\,(156 - 20)$ $\Rightarrow a^3 + b^3 = 1904$.
4.	2	$z \# x = z^2 - x^2$. $P \# 12 = 1012$ $\therefore P^2 - 12^2 = 1012. \Rightarrow P^2 = 1156 \Rightarrow P = 34$.
5.	3	$a + b + c = 3$, $a^2 + b^2 + c^2 = 6$, $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$ $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$, $(a + b + c)^2 = a^2 + b^2 + c^2 + 2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \times abc$, $3^2 = 6 + 2\,(1) \times abc$; $abc = \frac{3}{2}$
6.	2	Apply the hit and trial to find that the first part is the square of $(\sqrt{2} + 1)$. Similarly second part is the square of $(3 - \sqrt{2})$. After these two parts are added, it becomes 4, the square root of which is 2 and this is the answer.
7.	4	$2^{x+1}.5^x = 200$. Options can be tried. 5th option is the correct answer. Because $2^3.\ 5^2 = 8 \times 25 = 200$.
8.	1	Sum of roots $\Rightarrow x_1 + x_2 = \frac{-b}{a}$. Product of roots $= x_1x_2 = \frac{c}{a}$ Required sum of roots $= \frac{1}{x_1} + \frac{1}{x_2} = \frac{x_1 + x_2}{x_1x_2} = \frac{-b}{a} \times \frac{a}{c} = \frac{-b}{c}$ Product of roots $= \frac{1}{x_1} \times \frac{1}{x_2} = \frac{a}{c}$ Hence the required equation is $x^2 + \frac{bx}{c} + \frac{a}{c} = 0$ $\Rightarrow \therefore cx^2 + bx + a = 0$. Hence 1st option is the correct answer.
9.	1	Option three can be eliminated as 24 is not twice of 8. Going by options if Shriya's age is 10, Arjun's age would be 20. 5 years ago their ages would have been 5 yrs and 15 yrs. Hence three times.
10.	2	The question asks the square of the smaller number +1. So going by options we first subtract 1 from each and see the perfect squares. 50 – 1 = 49 which is square of 7. If 7 is the smaller number, then its 5 times is 35. If we add 21 to 7 and 35, they become 28 and 56. Also, 28 × 2 = 56. Hence 2nd option.